LIONEL PIGOU, dr. ir.

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EXPERIENCE OVERVIEW

Machine Learning: 6 years of experience
- 1 year in the biotech industry: graph neural networks (GNNs), CNNs, LLMs, DNA Transformers, MLOps.
- 4 years Ph.D. + 1 year master's thesis: deep neural networks in computer vision and NLP.
- Skills: PyTorch, Tensorflow, OpenCV, NumPy, Linux, git, Scikit-Learn, Python, scientific writing, MLFlow, GCP.

Software Engineering: 5 years of experience
- 5 years game development: graphics engineering, multiplayer, GPU compute, multithreaded job system.
- Internship full stack web application development.
- Skills: C#, Unity, Blender, (compute) shaders, GPGPU, JavaScript, C/C++, Java, Ruby on Rails.

PROFESSIONAL SUMMARY

Senior Machine Learning Expert
Oct 2023 - Present
BioLizard
Ghent, Belgium

Data science R&D, ML consultancy and software engineering in the biotech industry.
- Multi-species **graph neural networks** (GNNs) with multi-omics data, predicting gene-disease associations: full-stack development: **MLOps**, ML engineering, cloud compute, training strategy, experiment tracking.
- Predicting analyte concentrations in Raman spectra: **XGBoost**, 1D convolutional neural networks (CNNs), hyperparameter Bayesian optimization, data gathering consultancy.
- **Product engineering**: LLM-based literature discovery and text mining: extracting business analytics from thousands of papers. Co-lead in design, development and production.

Indie Game Developer
Sept 2018 - Oct 2023
Nel Stuff
Ghent, Belgium

Solo developing PC video games using Unity and publishing on Steam.
- Mad Adventures (2021 - 2022): a **multiplayer** party game with network-synchronized objects.
  - 177K unique players, 560 daily players**,** the **reviews are 93% positive** as of writing.
- Meor (2018 - 2021): an experimental procedurally generated voxel sandbox game on GPU.
  - The engine can draw an area representing **400 million dynamic voxels** on screen all at once.
  - The **multithreaded** job system and the **GPU compute** kernels make the game run smoothly.

Ph.D. in Deep Learning
Sept 2014 - Aug 2018
Ghent University
Ghent, Belgium

Deep learning research applied to video data: gesture and sign language recognition.
- **Awarded $100K for winning** an international Kaggle competition on classifying plankton images.
- State-of-the-art gesture recognition results: **90.6% label accuracy** using recurrent neural networks to capture the temporal structure in video. One of the two corresponding articles was **cited 578 times**.
- Coached ~10 master thesis students on various machine learning and deep learning topics.
- Responsible for the lab's computer servers and GPU cluster, including hardware assembly and software setup.
- **Published 2 papers** in high impact scientific journals and participated in 4 international conferences.

Full Stack Software Engineering Internship
Summer 2013
Solvace
Ghent, Belgium

EDUCATION

Ph.D. in Computer Science: Deep Learning (see above)
2014-2018
Ghent University

Master of Science in Computer Science Engineering: ICT
2012-2014
Ghent University

Master's thesis on deep learning. Won **5th place** in an international competition and was **cited 561 times**.